

Maintenance Manual — Nexus Semiconductor Fab 3

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Plant	Nexus Semiconductor Fab 3	Effective Date	15 Jan 2026
Document Owner	N. Holloway, Fab Equipment Engineering Manager	Classification	Controlled — Site Use

1. Purpose & Scope

This manual defines maintenance of process tools (lithography scanners, plasma etchers, deposition chambers, implanters), facility chillers, UPW skids, and gas/chemical distribution at Nexus Fab 3. Maintenance is segmented into PM, scheduled clean, and corrective; PM frequencies are wafer-count-driven.

2. Critical Asset Register

The table below lists the critical assets covered by this manual. Each asset has a detailed task package in the CMMS (SAP-PM / Maximo) linked to its asset ID.

Asset ID	Description	Maintenance Strategy
LITH-SCN-02	EUV scanner (Lithography)	Per-100k-wafer reticle stage clean; vendor PM per service contract.
ETCH-CHA-14	Plasma etcher chamber	Wet clean every 1000 RF hours; pressure decay test post-PM.
DEP-PVD-09	PVD deposition chamber	Shield kit swap at 800 wafers; target consumption tracked.
IMP-04	Ion implanter	Source life monitoring; beam tune verification after every PM.
UPW-SKID-01	Ultra-pure water skid	TOC, resistivity continuous; ion-exchange resin per 18-month cycle.

3. PM Schedule Summary

Frequency	Activity
Per wafer-count	Wet cleans on ETCH-CHA-14 and DEP-PVD-09.
Daily	UPW quality dashboard review; gas cabinet purge cycle audit.
Weekly	Particle baseline on representative chambers.
Monthly	FMEA review for top defect contributors.
Quarterly	Vendor PM audits on EUV scanner LITH-SCN-02.

4. Work Management

All maintenance work is authorized through a CMMS work order. Work orders are categorized as PM, PdM, Corrective, Emergency, or Project. Maintenance KPIs (Schedule Compliance ≥85%, PM Compliance ≥95%, MTBF) are reported weekly to the Plant Manager.

5. Procedures

5.1 Pass-Down and Permit

PM start requires a Tool Permit issued by Equipment Engineering with sign-off from Process Integration. Pass-down notes captured in the MES tool history.

5.2 Post-PM Qualification

After any PM, the tool runs a Particle Baseline (PB) and a Process Control Wafer (PCW). Tool returned to production only on PB and PCW passing the SPC criteria.

5.3 Spare Parts

Critical spares maintained at a 95% fill rate by the Fab Stores. Source life parts for implanter tracked individually with predicted-end-of-life alerts at 80% of expected lifetime.

6. References

- SEMI E10 — Equipment Reliability, Availability, and Maintainability
- SEMI E35 — Cost of Ownership
- Internal: PMM-NXS-EQ-01, PB-PCW-NXS-12

7. Revision History

Rev	Date	Author	Summary
1	04 Jun 2023	N. Holloway	Initial issue.
2	21 Nov 2024	N. Holloway	Added predictive analytics integration.
3	09 Jul 2025	N. Holloway	Updated asset register.
4	15 Jan 2026	N. Holloway	Refreshed PM frequencies and KPIs.

Approvals

Role	Name	Signature	Date
Maintenance Engineering	N. Holloway, Fab Equipment	<u>Engineering Manager</u>	15 Jan 2026
Reliability Engineering	Sr. Reliability Engineer	_____	15 Jan 2026
Operations / Site Lead	Q. Sandberg, Fab GM	_____	15 Jan 2026